

Database Management System

BBA — 2nd Semester — 2024 — Answer Sheet

Subject Code: N/A

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

Brief Answer Questions: [10 × 1 = 10]

1. Define Database Language.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Write a SQL statement for update operation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is unary Relationship?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. List any four types of attributes in ER Model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define serializable schedule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is big data?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define deferred data update.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is importance of creating view?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. State 3NF.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define Roles.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Short Answer Questions: (Attempt any FIVE Questions)[$5 \times 3 = 15$]

11. Describe shadow paging concept in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Who is DBA? List the roles of DBA.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. Explain the concept of Time stamp ordering concurrency control techniques.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. Illustrate how composite attribute is reduced into relational schema with example?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. Explain constraints of generalization and specialization in ER model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. What is the importance of database security?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section C

Long Answer Questions: (Attempt any THREE Questions)[$3 \times 5 = 15$]

17. Explain any four types of Database user.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

18. How normalization helps to design good database? Explain second normal form with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. Discuss any four constraints available in database with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. Explain the Desirable Properties of Transaction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section D

Comprehensive Answer / Case / Situation Analysis Questions:[2 × 10 = 20]

21. Draw an ER diagram for Hospital management system with at least five entity set. Implement the concept of mapping cardinality and extended ER feature. Assume suitable attribute sets for each entity set.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

22. Consider following relational database: Car (reg_no, model, price, color) Person (pid, name, address, phone) owns_by (reg_no, pid, date) Write SQL statement for the following: i) Insert a tuple {1, Ram Bdr, kathmandu} in person relation. ii) Display the model number with highest price. iii) Increase the price of each car by 15%. iv) Find name of person who owns red colored car. v) Display those records of car which model contains at most 4 characters.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.